

Yaswitha Ramisetty

AI / Machine Learning Engineer | Data Scientist | Generative AI | Agentic AI

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PROFILE SUMMARY

AI/ML Engineer and Data Scientist with 5+ years of experience building and deploying production machine learning, Generative AI, and data-driven solutions across banking, healthcare, and enterprise environments. Experienced in developing predictive models, Retrieval-Augmented Generation (RAG) applications, LLM-powered agents, multi-agent workflows, and scalable ML pipelines. Proven experience improving model accuracy, reducing data processing time, and automating document-intensive workflows across datasets containing millions of records and terabytes of data. Strong hands-on background in Python, PyTorch, Scikit-learn, XGBoost, LangChain, LangGraph, FastAPI, Apache Spark, Docker, Kubernetes, MLOps/LLMOps, and cloud AI services across AWS, Azure, and GCP.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, Java, C#, JavaScript, TypeScript, Bash/Shell Scripting

AI/ML & Data Science: Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Natural Language Processing (NLP), Computer Vision, Time Series Forecasting, Reinforcement Learning, Statistical Modeling, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Predictive Analytics

ML Frameworks & Libraries: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Hugging Face Transformers, Pandas, NumPy

Generative AI & Agentic AI: OpenAI, Azure OpenAI, LangChain, LangGraph, LlamaIndex, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Fine-Tuning, PEFT/LoRA, AI Agents, Agentic AI, Multi-Agent Systems, LLM Orchestration, LLM Evaluation, Tool Calling, Function Calling

Vector Databases & Retrieval: Azure AI Search, Qdrant, FAISS, Pinecone, ChromaDB, Vector Search, Semantic Search, Hybrid Search, BM25, Cross-Encoder Reranking, Embeddings, Chunking Strategies

Data Engineering & Big Data: Apache Spark, PySpark, Databricks, Kafka, Hadoop, Apache Airflow, ETL/ELT Pipelines, Data Warehousing, Batch Processing, Real-Time Data Processing, Event-Driven Architecture

MLOps, LLMOps & Deployment: MLflow, Kubeflow, Weights & Biases, Docker, Kubernetes, FastAPI, REST APIs, Microservices, CI/CD Pipelines, Model Monitoring, LLM Monitoring, Model Registry, Feature Store, GitHub Actions, Jenkins

Cloud Platforms: AWS (SageMaker, Bedrock, S3, EC2, Lambda, API Gateway), Google Cloud Platform / GCP (Vertex AI, Vertex AI Pipelines, BigQuery), Microsoft Azure (Azure AI Foundry, Azure OpenAI, Azure Machine Learning, Azure AI Search, Azure Functions, Azure Kubernetes Service (AKS), Azure Blob Storage, Azure Monitor)

Frontend & AI Applications: React.js, TypeScript, JavaScript, HTML5, CSS3, Gradio, REST API Integration, Asynchronous Programming

Databases & Storage: PostgreSQL, MySQL, MongoDB, Redis, NoSQL Databases

Responsible AI & Analytics: SHAP, LIME, Explainable AI (XAI), Responsible AI, AI Governance, Power BI, Tableau

EMPLOYMENT HISTORY

Comerica, Dallas, Texas, USA

Jan 2026 to Present

AI / Machine Learning Engineer

- Developed fraud detection, risk scoring, and financial forecasting models using Python, XGBoost, LightGBM, PyTorch, and Scikit-learn across **20M+ banking transactions**, improving **AUC-ROC by 15 points** and strengthening data-driven risk assessment.
- Engineered scalable ETL and feature engineering pipelines using Apache Spark, PySpark, Airflow, PostgreSQL, and MongoDB to process **10TB+ of structured and unstructured financial data**, reducing processing time by **40%**.
- Built automated ML pipelines covering feature engineering, hyperparameter optimization, model validation, and deployment, improving **F1-score by 18%** while accelerating model delivery.
- Fine-tuned domain-specific language models using Hugging Face Transformers, PEFT/LoRA, BERT, GPT, and spaCy for financial document intelligence and compliance workflows, improving classification performance and reducing dependency on manual document review.
- Designed enterprise RAG applications using OpenAI, LangChain, LangGraph, FAISS, and Pinecone to provide semantic search and grounded question answering across **500K+ financial documents**.
- Developed AI agents and multi-agent workflows using LangGraph and exposed model capabilities through FastAPI services deployed on AWS SageMaker, EC2, S3, and Lambda to automate customer support and operational workflows.
- Established production MLOps and LLMOps workflows using MLflow, Kubeflow, Docker, Kubernetes, CI/CD, model registries, monitoring, SHAP, and LIME to improve deployment consistency, model explainability, and governance.

Dell, Round Rock, Texas, USA

Feb 2025 to Dec 2025

Machine Learning Engineer

- Developed forecasting and optimization models using Python, XGBoost, LightGBM, PyTorch, TensorFlow, and reinforcement learning, improving **forecasting accuracy by 20%** for global supply-chain planning.
- Engineered large-scale data pipelines using Apache Spark, PySpark, Databricks, Kafka, and Hadoop to process **15TB+ of supply-chain data**, reducing analytics processing time by **45%**.
- Built end-to-end machine learning workflows on GCP using Vertex AI and BigQuery for data preparation, feature extraction, model training, experiment tracking, versioning, and endpoint deployment.
- Deployed scalable model training and real-time inference workloads using AWS SageMaker and S3, supporting model artifact management and production ML operations.
- Supported machine learning experimentation and deployment through Azure Machine Learning and Azure Blob Storage within a multi-cloud ML environment.
- Developed enterprise RAG applications using OpenAI, LangChain, FAISS, Pinecone, vector search, and prompt engineering to deliver citation-grounded information retrieval across **250K+ enterprise documents**.

- Designed LangGraph-based AI agents and multi-agent workflows for technical support automation, reducing manual resolution effort and improving response efficiency.
- Implemented production MLOps/LLMOps workflows using MLflow, Kubeflow, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, model monitoring, and automated retraining to support reliable model lifecycle management.

South Indian Bank, Hyderabad, India

Apr 2023 to Jun 2024

Data Scientist

- Analyzed **50M+ customer and transaction records** using Python, SQL, Apache Spark, and Hadoop to deliver risk, lending, portfolio, and customer analytics supporting faster business decision-making.
- Developed fraud detection, loan default prediction, and customer segmentation models using Scikit-learn, XGBoost, LightGBM, and TensorFlow, improving **prediction accuracy by 18%** and increasing **campaign conversion rates by 15%**.
- Built NLP-driven document intelligence and compliance analytics solutions using BERT, Transformers, spaCy, and NLTK, reducing **manual review effort by 45%** while improving document-processing efficiency.
- Designed scalable feature engineering, model validation, and analytics workflows that shortened **model development cycles by 30%**.
- Automated operational reporting with Power BI and Tableau, reducing **reporting effort by 40%** and manual operational processes by **25%** while improving access to real-time business insights.

GeBBS Healthcare Solutions, Hyderabad, India

Jan 2021 to Mar 2023

Data Scientist

- Developed predictive analytics and machine learning solutions using Python, SQL, Scikit-learn, and XGBoost across 20M+ EHR, claims, provider, and patient records, improving risk prediction accuracy by 15% and supporting initiatives that reduced hospital readmissions and improved patient care outcomes.
- Built healthcare analytics platforms for Revenue Cycle Management (RCM), denial prevention, and claims optimization, reducing claim processing errors by 20%, improving reimbursement efficiency, and accelerating revenue recovery across 8+ healthcare programs.
- Developed NLP-powered clinical intelligence applications using BERT, Transformers, spaCy, and NLTK to extract insights from physician notes, discharge summaries, and medical records, reducing manual review effort by 45% and improving clinical documentation and compliance workflows.
- Architected scalable healthcare data pipelines using Apache Spark, Hadoop, Airflow, and ETL frameworks, processing millions of patient records while reducing data processing time by 35% and enabling faster access to actionable clinical and operational insights.
- Delivered HIPAA-compliant analytics, KPI reporting, and executive dashboards using Power BI, Tableau, Pandas, and SQL across 15+ healthcare facilities, reducing reporting effort by 40%, improving regulatory reporting accuracy, and driving data-driven decisions for clinical, financial, and operational stakeholders.

PROJECTS

Agentic AI Document Intelligence Platform using RAG and LangGraph

Dec 2025 to Apr 2026

- Engineered a fully local, privacy-preserving RAG platform using Sentence Transformers (MiniLM), Qdrant vector search, GPT4All (LLaMA-3), and LangGraph for offline document intelligence over 100+ pages of research content.
- Architected a scalable retrieval pipeline combining dense vector search, lexical matching (RapidFuzz), and LRU caching, improving retrieval relevance by 25% while reducing redundant computation and inference latency.
- Built asynchronous FastAPI microservices with modular service orchestration and fault-tolerant request handling; productionized via Dockerized deployment with structured logging, monitoring hooks, and real-time multi-turn conversational workflows.
- Developed RESTful APIs and React + Gradio interfaces for interactive document querying and conversational research assistance.

Explainable Multimodal Alzheimer's Disease Prediction using Ensemble Machine Learning (ADNI)

Sep 2024 to Jan 2025

- Developed an end-to-end machine learning pipeline using the ADNI Alzheimer's Disease dataset, integrating MRI, PET, CSF biomarkers, APOE genotype, and clinical data through preprocessing, feature engineering, and data fusion for cognitive decline prediction.
- Built and optimized Random Forest, XGBoost, and ensemble classification models using Scikit-learn and XGBoost, leveraging stratified validation and hyperparameter tuning to improve prediction performance and model robustness.
- Engineered clinically relevant biomarkers, handled missing data, encoded categorical variables, and implemented feature selection to improve model generalization and support reliable disease classification.
- Applied SHAP-based Explainable AI (XAI) to interpret model predictions, identify key Alzheimer's risk factors, and provide transparent, clinically interpretable insights for decision support.

EDUCATION

Texas Tech University, Lubbock, TX

Master of Science in Computer Science

CERTIFICATIONS

- Google Cloud Professional Machine Learning Engineer
- IBM AI Engineering Professional Certificate
- DeepLearning.AI Deep Learning Specialization
- AWS Certified Cloud Practitioner
- Udemy AI Engineer Core Certification